

LESSON 2-5 • SESSION 1 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find Q1 and Q3

Builds on Lesson 2-4 Session 1 • 6.SP.B.4

TODAY'S LESSON

Camp hikes give seven distances. You need Q1 and Q3 before you find the IQR.

YOU NEED TOFind the middle of the lower half and of the upper half. **You will use it in Practice Form A, problems 1 and 3.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** **Order** the data and find the median.
- 2** Split into a **lower** half and an upper half.
- 3** Q1 = middle of the lower half. **Q3** = middle of the upper.

WATCH ME*the steps in action***Find Q1 and Q3 for 12, 5, 20, 9, 15, 25, 18.****Step 1** 5, 9, 12, 15, 18, 20, 25. The median is 15.**Step 2** Lower half: 5, 9, 12. Upper half: 18, 20, 25.**Step 3** Q1 = 9. Q3 = 20.**Answer Q1 = 9, Q3 = 20****TRY IT WITH HELP***fill in each step*

- 1.** Find Q1 and Q3 for 2, 4, 5, 8, 11, 13, 16.

Step 1 The median is _____.**Step 2** Lower half: _____.**Step 3** Q1 = _____. Q3 = _____.

- 2.** Find Q1 and Q3 for 10, 15, 20, 25, 30, 35.

Step 1 The median is _____.**Step 2** Lower half: _____.**Step 3** Q1 = _____. Q3 = _____.**YOUR TURN***use the steps on your own*

- 3.** Each list is in order. Write Q1 and Q3.

a. 3, 6, 9, 12, 15, 18, 21, 24 _____**b.** 1, 1, 2, 4, 6, 7 _____**c.** 20, 24, 26, 31, 33 _____**d.** 41, 44, 46, 50, 53, 57, 60, 62, 65 _____

Read Box Plot Values

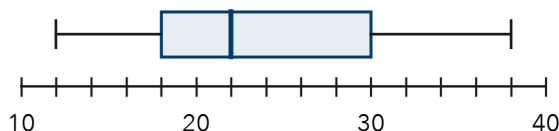
Builds on Lesson 2-4 Session 2 • 6.SP.B.4

TODAY'S LESSON

In problem 8 you read two box plots to find each snack bar's range and IQR.

YOU NEED TORead min, Q1, Q3, and max from a box plot, then subtract. **You will use it in Practice Form A, problems 7 and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Read the whisker **ends**: min and max.
- 2** Read the box **edges**: Q1 and Q3.
- 3** **Subtract**: range = max – min. IQR = Q3 – Q1.

WATCH ME*the steps in action***Find the range and the IQR.****Step 1** Min 12, max 38.**Step 2** Q1 18, Q3 30.**Step 3** Range $38 - 12 = 26$. IQR $30 - 18 = 12$.**Answer** range **26**, IQR **12****TRY IT WITH HELP***fill in each step*

- 4.** Whiskers go from 10 to 45. The box goes from 20 to 35.

Step 1 Min = _____. Max = _____.**Step 2** Q1 = _____. Q3 = _____.**Step 3** Range = _____. IQR = _____.

- 5.** Whiskers go from 104 to 140. The box goes from 116 to 132.

Step 1 Min = _____. Max = _____.**Step 2** Q1 = _____. Q3 = _____.**Step 3** Range = _____. IQR = _____.**YOUR TURN***use the steps on your own*

- 6.** Find the range and the IQR.

a. Min 5, Q1 10, Q3 20, Max 30: ____**b.** Min 12, Q1 20, Q3 26, Max 41: ____**c.** Min 0, Q1 15, Q3 40, Max 60: ____**d.** Min 50, Q1 58, Q3 71, Max 90: ____**READY CHECK***a problem like the one in today's lesson*

- 7.** A box plot shows 11, 14, 16, 19, 62. What is the IQR?

A 51**B** 5**C** 3**D** 16**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #____